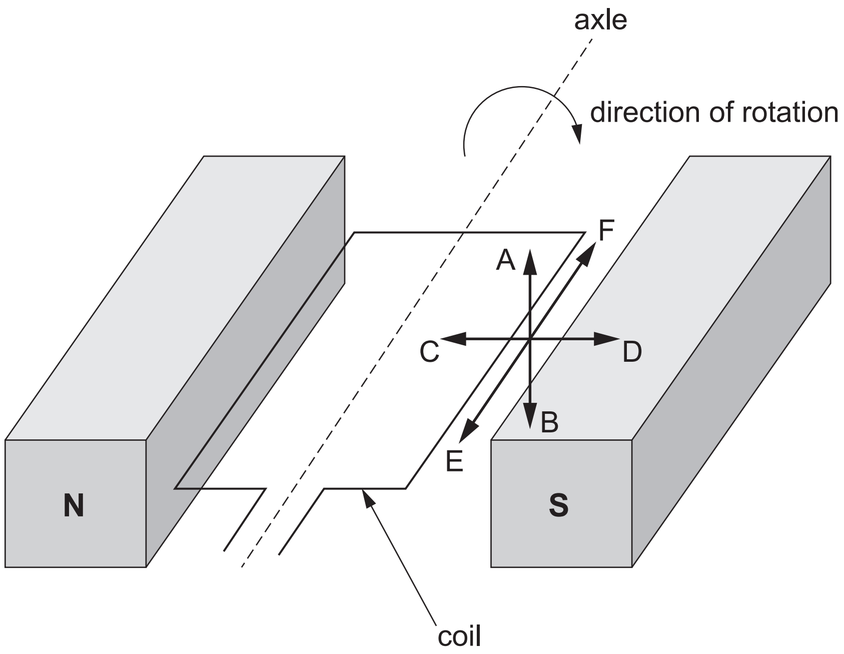
Examiner only

**5.** The operation of electric generators can be explained by electromagnetic induction.

(a) The diagram shows part of an a.c. generator.



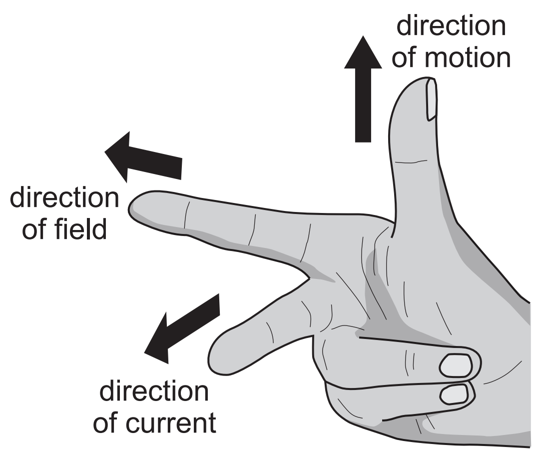

(i) State the direction (A, B, C, D, E or F) in which the magnetic field acts. [1]

Direction = ......................

(ii) State the direction (A, B, C, D, E or F) in which the right hand side of the coil moves. [1]

Direction = ......................

(iii) Use Fleming's right hand rule to determine the direction (A, B, C, D, E or F) of the induced current in the right hand side of the coil. [1]




Direction = ......................

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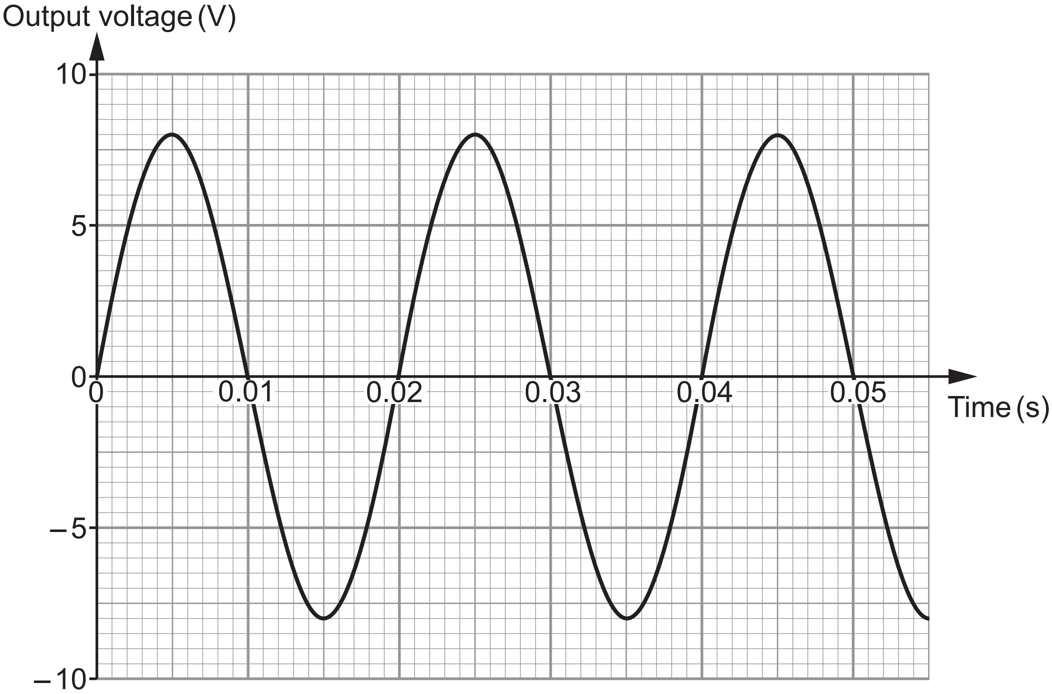
(iv) When the coil spins, a voltage is produced.
This is shown on the graph.




I. Use the graph to find the maximum voltage produced by the generator. [1]

maximum voltage = ............... V

II. Use the graph to find the time for one rotation (spin) of the coil. [1]

time = ................. s

III. **Complete the table** below to show the effect of each change made.
Use the words **increases**, **decreases** or **no effect**.
Three boxes have been completed for you. [3]

| Change made | Effect on maximum voltage | Effect on time for 1 rotation (spin) |
|---|---|---|
| Stronger magnets | increases | no effect |
| More turns on the coil | .................................... | no effect |
| Coil spins slower | .................................... | .................................... |

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(b) Transformers also work by electromagnetic induction.

(i) Tick (✓) the boxes next to the **three** correct statements about transformers. [3]

| | |
|---|---|
| Transformers change a.c. to d.c. | ☐ |
| Transformers change d.c. voltages. | ☐ |
| The core is made from iron. | ☐ |
| There is an electric current in the core. | ☐ |
| The primary coil creates an alternating magnetic field. | ☐ |
| Transformers can increase or decrease voltages. | ☐ |

(ii) A transformer increases voltage from 12 V to 36 V.
The primary coil contains 20 turns of wire.
Use the equation:

$$N_2 = N_1 \times \frac{V_2}{V_1}$$

to calculate the number of turns in the secondary coil, $N_2$. [2]

$N_2$ = .........................

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